Izmir Institute of Technology

Electrical and Electronics Eng. Dep.

EE 471 Modern Software Development Practices and Technologies

In-class exercise Week #5

Thursday 26/10/2026

Preliminary Task: Study the shared course materials [1, 2, 3, 4, 5, 6]

Task 1: Leverage vibe coding to create a speech-studio-backed website

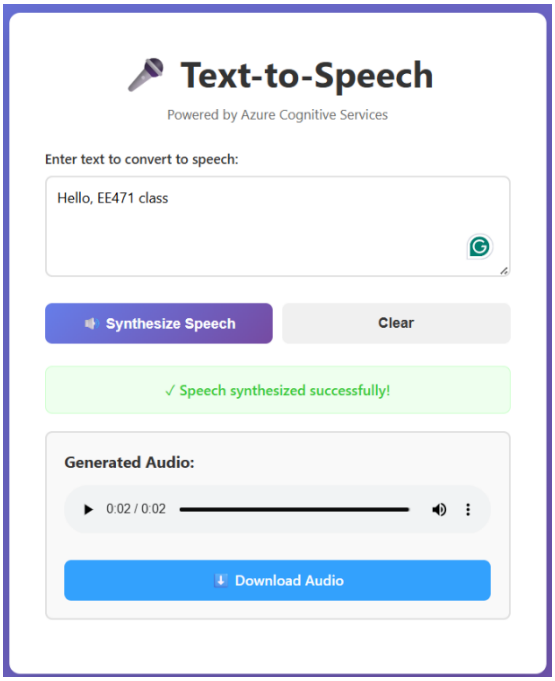
You are working in a startup, and your customer asked your team to develop a website that a user can translate:

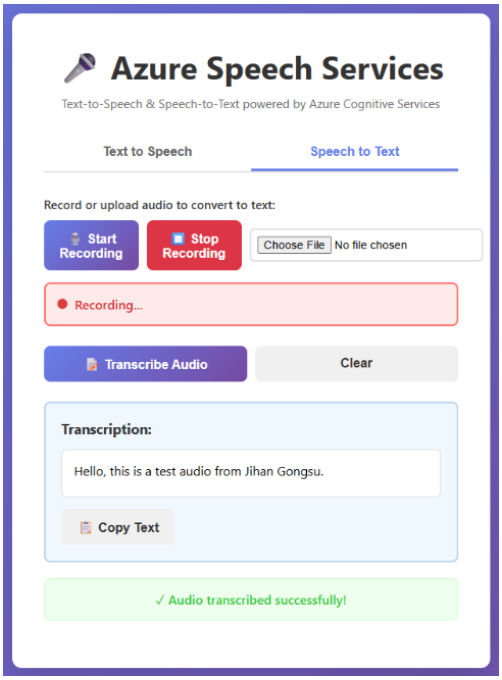
- text-to-speech and
- speech to text

Your team lead wants to deliver it ASAP, and you decided work with a colleague of yours. As a team, one developer works on text-to-speech, and the other is on speech to text.

PART 0 (STRONGLY RECOMMENDED):

Create a .env file to keep your secret keys, “e.g. Azure SPEECH_KEY”, and use “python-dotenv” library to use it in your scripts. Update your “.gitignore” so that “.env” file is not pushed to remote and stays in your local **as they are sensitive information**.





Part 1: Text to speech

1) In your github repository, create a folder for Week-5-Task-1
2) Open a branch with the name “text-to-speech”
3) Develop a website running on your localhost (with a Flask-based backend). This should be capable of translating texts entered by the users to an audio file.
4) Push it to your branch, and merge it to main.

Part 2: Speech to text

1) In your github repository open a branch with the name “speech-to-text”
2) Develop a website running on your localhost (with a Flask-based backend). This should be capable of translating audio files / or speech inputs given by the users into a text script.
3) Push it to your branch, and merge it into main.

Part 3: Resolve conflicts / Update your script for both functions

1) In Part 2, merging will fail and raise merge conflicts.
2) Resolve the conflicts, and update your code so that your new website has both of the features: text-to-speech & speech-to-text

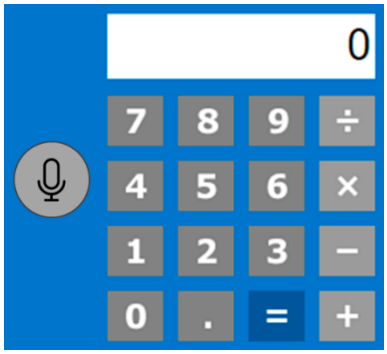
Part 4: Extend with the rest of the supplementary materials

1) Improve your script with real-time diarization, speech translation, and pronunciation assessment

Deliverables:

1) Your git repository link
2) “Task-1-demo.avi” - A screen recording (with audio) demo of your smart calculator with all functions.
3) “Task-1-code-explained.avi” - A screen recording (with audio) of you explaining each code block in your scripts.

Task 2: Voice-activated calculator



1) In your github repository, create a folder for Week-5-Task-2
2) Improve the calculator you designed in Mini-project 1 and Week #4 in-class exercise so that your calculator will run as a voice-activated calculator. **Microphone button** on the left will let users give voice commands to the calculator, and the calculator will execute them (Button→Voice command).

Command table:

0	1	2	3	4	5	6	7	8	9
Zero	One	Two	Three	Four	Five	Six	Seven	Eight	Nine
.	÷	x	+	-	=				
Point	Over	Times	Plus	Minus	Is				

3) The rest of the buttons are also magical, and they let the **calculator speak** in line with the commands above.
   Example voice output can be “Three”, “plus”, “five,” “is eight” is spoken after the buttons “3”, “+”, “5”, and “=” are pressed. Or respectively,
4) Push your scripts to Github.

Deliverables:

1) Your git repository link
2) “Task-2-demo-1.avi” - A screen recording (with audio) of a demonstration of your smart calculator with the following **audio inputs**:
   a. “Five”, “plus”, “two”, “is”
   b. “Seven”, “minus”, “three”, “is”
   c. “Three”, “times”, “two”, “is”
   d. “Nine”, “over,”, “three”, “is”

3) “Task-2-demo-2.avi” - A screen recording (with audio) of a demonstration of your smart calculator with the following **button press inputs**:
    a. “5”, “+”, “2, “=”
    b. “7”, “-”, “3, “=”
    c. “3”, “x”, “2”, “=”
    d. “9”, “÷“, ”3“, “=”
4) Your vibe coding transcripts.
5) “Task-2-code-explained.avi” - A screen recording (with audio) of you explaining each code block in your scripts.